

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460174.9422 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.01 +/- 0.049
Transit depth (Rp/Rs)^2	0.0094 +/- 0.094 [%]
Orbital Inclination (inc)	86.75 +/- 1.42
Airmass coefficient 1 (a1)	0.81 +/- 0.24
Airmass coefficient 2 (a2)	0.21 +/- 1.13
Scatter in the residuals of the lightcurve fit is	32.08 %
Best Comparison Star	#1 - [158, 422]
Optimal Aperture	3.0
Optimal Annulus	3.5
Transit Duration (day)	0.0471 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.01 ± 0.049 vs 0.1301 (deviation 2.2σ)
Duration fit vs published	0.0471 d vs 0.0908 d (deviation 16.32σ)
Mid-time O–C	0.1 min
Depth SNR	0.2
Residual scatter	32.08 %

Light curve

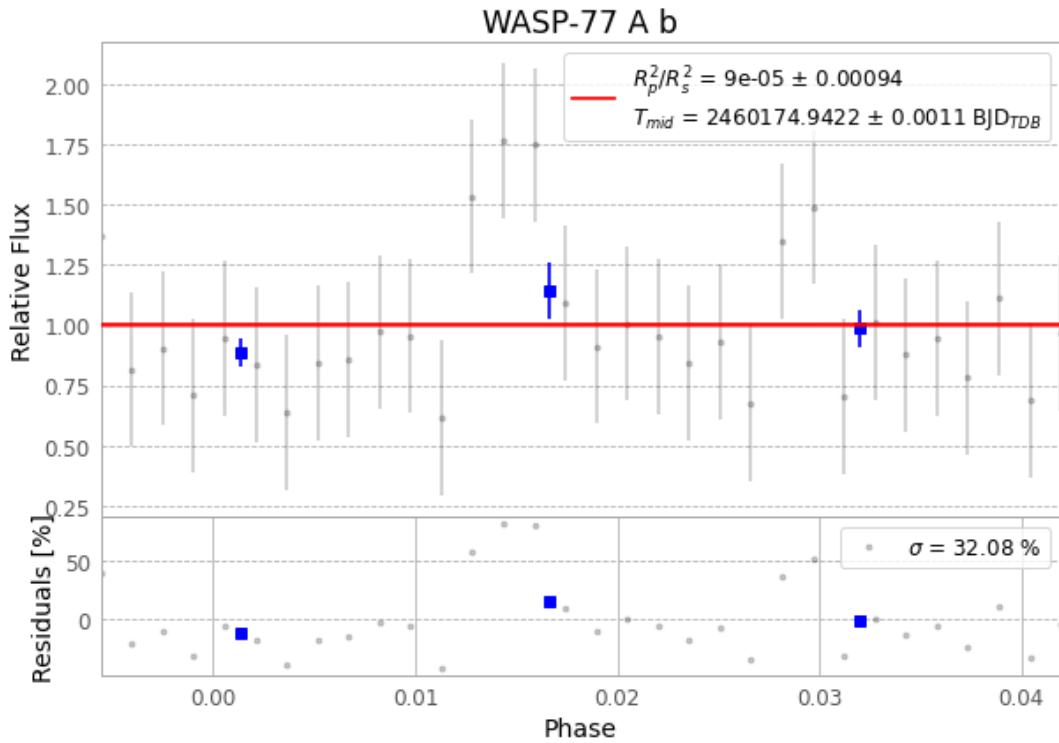


Figure 1 — FinalLightCurve_WASP-77 A b_2023-08-18.png (SHA-256 1ccb008ae207b002...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [552, 371], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7b813dc16e58b1a5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

32 FITS frames (21 MB), WASP-77230818102115.FITS → WASP-77230818115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-230818.log (4f4e6c38a76e...), FinalLightCurve_WASP-77 A b_2023-08-18.png (1ccb008ae207...), FinalLightCurve_WASP-77 A b_2023-08-18.csv (9ba67fa2c630...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 19:03Z.